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Register No.

Half-Yearly Examination - 2025

PHYSICS

Time : 3.00 Hrs. Marks : 70

PART - I

I. Answer all the questions. 15x1=15

Choose the correct answer.

1. Which of the following electro magnetic radiations is used for viewing objects through fog
   a) microwave b) gamma rays c) X-rays d) infrared
2. If voltage applied on a capacitor is increased from V to 2V, choose the correct conclusion.
   a) Q remains the same, C is doubled b) Q is doubled, C doubled
   c) C remains same, Q doubled d) Both Q and C remain same
3. The internal resistance of a 2.2V cell which gives a current of 0.2 A through a resistance of 10Ω is
   a) 0.2 Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω
4. A circular coil of radius 5 cm and 50 turns carries a current of 3 ampere. The magnetic dipole moment of the coil is nearly
   a) 1.0 A m^2 b) 1.2A m^2 c) 0.5A m^2 d) 0.8A m^2
5. An object is placed in front of a convex mirror of focal length of f and the maximum and minimum distance of an object from the mirror such that the image formed is real and magnified.
   a) 2f and c b) c and ∞ c) f and O d) none of these
6. The flux linked with a coil at any instant **t** is given by $\Phi_B = 10t^2 - 50t + 250$. The induced emf at t = 3 s is
   a) -190V b) -10V c) 10 V d) 190 V
7. Light transmitted by Nicol prism is
   a) partially polarized b) unpolarized c) plane polarized d) elliptically polarized
8. The threshold wavelength for a metal surface whose photoelectric work function is 3.313 eV is
   a) 4125 Å b) 3750 Å c) 6000 Å d) 2062.5 Å
9. If the nuclear radius of ^{27}Al is 3.6 fermi, the approximate nuclear radius of ^{64}Cu in fermi is
   a) 2.4 b) 1.2 c) 4.8 d) 3.6
10. The zener diode is primarily used as
    a) Rectifier b) Amplifier c) Oscillator d) Voltage regulator
11. "Ski wax" is an application of nano product in the field of
    a) Medicine b) Textile c) Sports d) Automotive industry
12. The permittivity of free space or vacuum and its value is _____.
    a) 8.85 x $10^{-12}C^2N^{-1}m^{-1}$ b) 8.85 x $10^{-12}C^2N^{-1}m^{-2}$ c) 8.58 x $10^{-12}C^2N^{-1}m^{-2}$ d) 9.85 x $10^{9}C^2N^{-1}m^{-2}$
13. The SI unit of magnetic flux is _____. a) Web^2 b) Web m^2 c) T m^2 d) Web^{-2}
14. When light of wavelength 2200 Å falls on Cu, photo electrons are emitted from it. Find the threshold wavelength _____(The work function for Cu is ϕ_0 =4.65 eV)
    a) 2672 Å b) 2762 Å c) 7622 Å d) 2276 Å
15. The term _____ is called numerical aperture NA.
    a) n cosβ b) n sinβ c) sin (a + b) d) 2n sinβ

PART - II

II. Answer any six of the following questions. Question No.24 is compulsory. 6x2=12

16. State Fleming's left hand rule.

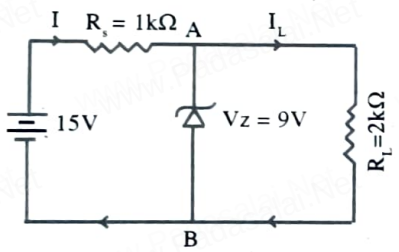
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17. Why steel is preferred in making Robots?
18. What does RADAR stand for?
19. Define 'capacitance'. Give its unit.
20. What is a photo cell? Mention the different types of photocells.
21. What is displacement current?
22. Calculate the power of the lens of the spectacles needed to rectify the defect of nearsightedness for a person who could see clearly up to a distance of 1.8m.
23. Define impact parameter.
24. Find the heat energy produced in a resistance of 10 Ω when 5A current flows through it for 5 minutes.

PART - III

III. Answer any six of the following questions. Question No.26 is compulsory. **6x3=18**

25. Write down any six properties of electromagnetic waves.
26. The repulsive force between two magnetic poles in air is 9×10^{-3}N. If the two poles are equal in strength and are separated by a distance of 10 cm, calculate the pole strength of each pole.
27. Obtain the expression for energy stored in the parallel plate capacitor.
28. Differentiate between Fresnel and Fraunhofer diffraction.
29. Explain the determination of the internal resistance of a cell using voltmeter.
30. List out the laws of photoelectric effect.
31. How will you induce an emf by changing the area enclosed by the coil?
32. Find the current through the Zener diode when the load resistance is 2 Ω. Use diode approximation.



33. Derive the relation between f and R for a spherical mirror.

PART - IV

IV. Answer all the questions in detail. **5x5=25**

34. a) Obtain the condition for bridge balance in Wheatstone's bridge.
    (OR)
    b) Explain about compound microscope and obtain the equation for the magnification.
35. a) Write down Maxwell equations in integral form.
    (OR)
    b) Derive the mirror equation and the equation for lateral magnification.
36. a) Derive an expression for electrostatic potential due to an electric dipole.
    (OR)
    b) Explain the construction and working of a full wave rectifier.
37. a) Find the magnetic field due to a long straight conductor using Ampere's Circuital Law.
    (OR)
    b) Obtain Einstein's photoelectric equation with necessary explanation.
38. a) Explain the construction and working of transformer.
    (OR)
    b) Discuss the spectral series of hydrogen atom.

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